Kunj Rathod

kunjrathod.com +1 (385) 202-8879 kunj.rathod@utah.edu linkedin.com/in/rathodkunj github.com/rathodkunj2005

Education

University of Utah Aug 2023 – Dec 2026
B.S. in Computer Science GPA: 3.7/4.0, Dean's List
Coursework: Distributed Systems, Operating Systems, Machine Learning, Computer Vision, NLP, Algorithms, Databases

Industry Experience

Software Engineering Intern — Microsoft Fabric (Azure Data) May 2026 – Aug 2026
Microsoft, Redmond, WA

- Building the SKU Change History feature end-to-end (query builder, service layer, REST API) for the Fabric Manage Capacities admin portal, integrating internal telemetry pipelines to give capacity admins auditable SKU-change timelines across large enterprise tenants.
- Migrating capacity overage workflows into native Fabric platform UX with action-driven side panes; defining data contracts with UX and backend teams across 3 time zones.

Software Development Intern, AI Services (SUDO Program) Jan 2025 – Present
University of Utah Health, Salt Lake City, UT

- Architected HIPAA-compliant serving infrastructure for 90+ hospital executives: token-streaming inference at p95 <200ms TTFT and 40% latency reduction via pipeline optimization and API caching.
- Owned API design, schema, and AWS infrastructure (Lambda, API Gateway, RDS, DynamoDB, S3, Bedrock); shipped 6 features across 4 sprints and implemented distributed session persistence for 1,000+ concurrent conversations.

AI Engineering Intern Nov 2024 – Apr 2025
CourtEasy.ai / Nugen (Remote)

- Scaled hybrid retrieval (dense vectors + BM25 + reranking) over 10M+ documents, improving recall 28% and reducing hallucinations 35%; benchmarked 8 model families on F1, latency, and cost and built routing that cut inference spend.

Selected Projects

BioGraphRAG — Distributed Biomedical Knowledge Graph System May 2024 – Jan 2025
Python, FastAPI, NebulaGraph, LlamaIndex, Docker, AWS

- Engineered a distributed GraphRAG system over 1M+ biomedical entities (UniProt, AlphaFold, RXNav) with a 40% accuracy gain over embedding-only baselines; optimized graph traversal 3x via caching and node pruning (p95 <500ms) and built ETL pipelines processing 2M+ entity updates monthly.

Continuum — Local-First Multimodal Memory Engine Jun 2026
Rust, Tauri 2, React, TypeScript, LanceDB, OCR, embeddings, MCP

- Built a privacy-first desktop memory layer capturing foreground context on-device and storing vector + lexical retrieval fields in LanceDB, exposing agent-ready recall through an MCP server.

HirePilot — Autonomous AI Recruiting Pipeline (Podium AI Hackathon Winner) Mar 2026
Python/FastAPI, Anthropic, Perplexity, GitHub REST, Twilio, Slack, Google Calendar

- Shipped a 9-agent recruiting pipeline in 24 hours integrating GitHub sourcing, Perplexity enrichment, live Twilio voice interviews, ranked shortlisting, and manager approval controls.

Research

MEMROT v2 — Mechanistic Evaluation of Long-Context Memory Failure May 2026 – Present
Independent; Gemma-2-2B, Gemma Scope SAEs, LongMemEval-S, CHPC/SLURM

- Built a reproducible mechanistic-evaluation pipeline (gold-position randomization, SAE layer scans, mean-ablation and intervention tests) running an 840-pass sweep over 140 controlled retrieval questions on distributed CHPC/SLURM compute.

Circuit-Level Generalization via Sparse Feature Circuits Jan 2026 – Present
University of Utah; attribution graphs, graph skeletonization

- Open-sourced a graph-skeletonization tool that prunes attribution graphs and groups nodes into supernodes via Mapper embeddings, exporting visualization-compatible artifacts without dropping source metadata.

Languages & Technologies

Languages: Python, C++, Java, Rust, Swift, TypeScript, SQL
Systems & Cloud: AWS (Lambda, API Gateway, S3, DynamoDB, RDS, Bedrock), Azure, Docker, Kubernetes, FastAPI, gRPC, CI/CD, SLURM, distributed systems
Data & AI: RAG/GraphRAG, vector search, BM25, reranking, NebulaGraph, LanceDB, LlamaIndex, PyTorch, TensorFlow, model evaluation